Lesson 26: Lagrangian Duality and KKT Conditions

AI/ML Mathematics

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1 Introduction

Duality theory transforms a constrained optimization problem into a related problem—the *dual*—whose optimal value provides a bound on the original (primal) problem. When the bound is tight (strong duality), the dual offers both computational advantages and deep structural insight: it reveals the sensitivity of the optimal value to perturbations in the constraints and leads directly to the Karush–Kuhn–Tucker (KKT) conditions, which are the central tool for characterizing optimal solutions of convex programs.

This lesson develops Lagrangian duality for the convex optimization problem in standard form:

$$\begin{aligned} \min_{x\in\mathbb{R}^n} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \le 0, \quad i = 1,\dots,m, \\ & a_j^\top x = b_j, \quad j = 1,\dots,p, \end{aligned} \tag{1}$$

where $f_0, f_1, \dots, f_m$ are convex and the equality constraints are affine. We denote the feasible set by $\mathcal{F}$ and the optimal value by p^*.

Section 2 introduces the Lagrangian and the dual function. Section 3 proves weak duality. Section 4 presents Slater's condition and proves strong duality. Section 5 derives the KKT conditions and proves both their necessity (under strong duality) and sufficiency (for convex problems). Section 6 works through detailed examples, and Section 7 provides exercises.

2 The Lagrangian and Dual Function

Definition 2.1 (Lagrangian)**.** The *Lagrangian* $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$ associated with (1) is

$$L(x,\lambda,\nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \nu_j (a_j^\top x - b_j), \tag{2}$$

where $\lambda = (\lambda_1,\dots,\lambda_m) \in \mathbb{R}^m$ are the *dual variables* (or *Lagrange multipliers*) for the inequality constraints and $\nu = (\nu_1,\dots,\nu_p) \in \mathbb{R}^p$ are the dual variables for the equality constraints.

Remark 2.2. The Lagrangian can be viewed as a "relaxation" of the primal problem. Instead of enforcing the constraints exactly, we penalize violations through the multipliers: a large λ_i imposes a heavy cost for violating $f_i(x) \le 0$, while ν_j penalizes deviations from the equality $a_j^\top x = b_j$ in either direction.

Definition 2.3 (Lagrange dual function)**.** The *Lagrange dual function* $g : \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R} \cup \{-\infty\}$ is defined by

$$g(\lambda,\nu) = \inf_{x\in\mathbb{R}^n} L(x,\lambda,\nu) = \inf_{x\in\mathbb{R}^n} \left\{ f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \nu_j (a_j^\top x - b_j) \right\}. \tag{3}$$

Proposition 2.4 (Dual function is concave)**.** *For any optimization problem (not necessarily convex), the Lagrange dual function g is concave.*

Proof. For each fixed $x \in \mathbb{R}^n$, the function $(\lambda,\nu) \mapsto L(x,\lambda,\nu)$ is affine in (λ,ν), because L is a sum of terms each linear in λ_i or ν_j. An affine function is both convex and concave. The pointwise infimum of a family of concave functions is concave:

Let $\alpha \in [0,1]$ and (λ,ν), (λ',ν') be given. For each x:

$$L(x, \alpha\lambda + (1-\alpha)\lambda', \alpha\nu + (1-\alpha)\nu') = \alpha L(x,\lambda,\nu) + (1-\alpha) L(x,\lambda',\nu').$$

Taking the infimum over x on both sides:

$$\begin{aligned} g(\alpha\lambda+(1-\alpha)\lambda', \alpha\nu+(1-\alpha)\nu') &= \inf_x \big[\alpha L(x,\lambda,\nu) + (1-\alpha)L(x,\lambda',\nu')\big] \\ &\geq \alpha \inf_x L(x,\lambda,\nu) + (1-\alpha)\inf_x L(x,\lambda',\nu') \\ &= \alpha\, g(\lambda,\nu) + (1-\alpha)\, g(\lambda',\nu'), \end{aligned}$$

where the inequality uses $\inf(f+g) \geq \inf f + \inf g$. This establishes concavity. □

Example 2.5 (Dual of a linear program)**.** Consider the LP: $\min c^\top x$ s.t. $Ax \leq b$, with $A \in \mathbb{R}^{m\times n}$. The Lagrangian is $L(x,\lambda) = c^\top x + \lambda^\top(Ax-b) = (c+A^\top\lambda)^\top x - \lambda^\top b$. The infimum over x is $-\infty$ unless $c + A^\top\lambda = 0$, so

$$g(\lambda) = \begin{cases} -\lambda^\top b & \text{if } A^\top\lambda + c = 0, \\ -\infty & \text{otherwise.} \end{cases}$$

The dual problem is $\max_{\lambda\geq 0}\{-\lambda^\top b : A^\top\lambda + c = 0\}$, which is the standard LP dual.

Example 2.6 (Dual of a QP)**.** Consider $\min \frac{1}{2}x^\top Px + q^\top x$ s.t. $Ax \leq b$ with $P \succ 0$. The Lagrangian is $L(x,\lambda) = \frac{1}{2}x^\top Px + (q+A^\top\lambda)^\top x - \lambda^\top b$. Setting $\nabla_x L = Px + q + A^\top\lambda = 0$ gives $x^*(\lambda) = -P^{-1}(q + A^\top\lambda)$. Substituting:

$$g(\lambda) = -\frac{1}{2}(q+A^\top\lambda)^\top P^{-1}(q+A^\top\lambda) - \lambda^\top b.$$

This is a concave quadratic in λ, so the dual is also a QP.

3 Weak Duality

Definition 3.1 (Dual problem)**.** The *Lagrange dual problem* associated with (1) is

$$\max_{\lambda\geq 0,\,\nu}\ g(\lambda,\nu). \tag{4}$$

Its optimal value is denoted d^*. A pair (λ,ν) with $\lambda \geq 0$ is called *dual feasible*. The quantity $p^* - d^*$ is the *duality gap*.

Theorem 3.2 (Weak duality)**.** *For any optimization problem (not necessarily convex),*

$$d^* \leq p^*. \tag{5}$$

Proof. Let $\tilde{x}$ be any primal feasible point: $f_i(\tilde{x}) \leq 0$ for $i = 1,\ldots,m$ and $a_j^\top\tilde{x} = b_j$ for $j = 1,\ldots,p$. Let (λ,ν) be any dual feasible pair: $\lambda \geq 0$.

We compute:

$$\begin{aligned} L(\tilde{x},\lambda,\nu) &= f_0(\tilde{x}) + \sum_{i=1}^m \underbrace{\lambda_i}_{\geq 0}\underbrace{f_i(\tilde{x})}_{\leq 0} + \sum_{j=1}^p \nu_j \underbrace{(a_j^\top\tilde{x} - b_j)}_{=0} \\ &\leq f_0(\tilde{x}). \end{aligned} \tag{6}$$

Taking the infimum over $x \in \mathbb{R}^n$:

$$g(\lambda,\nu) = \inf_x L(x,\lambda,\nu) \leq L(\tilde{x},\lambda,\nu) \leq f_0(\tilde{x}). \tag{7}$$

Since this holds for every primal feasible $\tilde{x}$:

$$g(\lambda,\nu) \leq \inf_{\tilde{x}\in\mathcal{F}} f_0(\tilde{x}) = p^*. \tag{8}$$

Since this holds for every dual feasible (λ,ν):

$$d^* = \sup_{\lambda\geq 0,\nu} g(\lambda,\nu) \leq p^*. \tag{9}$$

□

Weak Duality

For any optimization problem, $d^* \leq p^*$. The dual function $g(\lambda, \nu)$ evaluated at any dual-feasible point provides a *lower bound* on the primal optimal value. This holds regardless of convexity.

Corollary 3.3 (Certificate of optimality)**.** *If a primal feasible x and dual feasible (λ, ν) satisfy $f_0(x) = g(\lambda, \nu)$, then x is primal optimal and (λ, ν) is dual optimal.*

Proof. By weak duality, $g(\lambda, \nu) \leq d^* \leq p^* \leq f_0(x)$. If $f_0(x) = g(\lambda, \nu)$, all inequalities are equalities. □

4 Strong Duality and Slater's Condition

Definition 4.1 (Strong duality)**.** *Strong duality* holds for an optimization problem when $d^* = p^*$, i.e., the duality gap is zero.

Definition 4.2 (Slater's condition)**.** The convex program (1) satisfies *Slater's condition* (or *Slater's constraint qualification*) if there exists a point $\hat{x} \in \operatorname{relint}(\operatorname{dom} f_0)$ such that

$$f_i(\hat{x}) < 0, \quad i = 1, \ldots, m, \qquad a_j^\top \hat{x} = b_j, \quad j = 1, \ldots, p. \tag{10}$$

Such a point $\hat{x}$ is called a *strictly feasible point*. Note that strict inequality is required for the inequality constraints, but the equality constraints need only be satisfied exactly.

Remark 4.3. If some inequality constraints are affine (say f_i is affine for $i \in \mathcal{A}$), Slater's condition can be relaxed: the affine inequalities need only be satisfied with $\leq$ (not strictly $<$) at the Slater point. This is sometimes called the *refined Slater condition*.

Theorem 4.4 (Slater implies strong duality)**.** *If the convex program (1) satisfies Slater's condition and p^* is finite, then strong duality holds: $d^* = p^*$. Moreover, the dual optimal value is attained: there exist $\lambda^* \geq 0$, ν^* with $g(\lambda^*, \nu^*) = d^* = p^*$.*

Proof. We use a supporting hyperplane argument on the set of achievable constraint slack and objective values.

Step 1: Define the achievable set. Let

$$\mathcal{G} = \big\{(u, v, t) \in \mathbb{R}^m \times \mathbb{R}^p \times \mathbb{R} : \exists\, x \text{ with } f_i(x) \leq u_i, \;\; a_j^\top x - b_j = v_j, \;\; f_0(x) \leq t\big\}. \tag{11}$$

This set is convex. To see this, let (u, v, t) and (u', v', t') lie in $\mathcal{G}$, witnessed by x and x' respectively. For $\alpha \in [0, 1]$, set $\bar{x} = \alpha x + (1-\alpha)x'$. By convexity of f_i: $f_i(\bar{x}) \leq \alpha f_i(x) + (1-\alpha) f_i(x') \leq \alpha u_i + (1-\alpha) u_i'$. Similarly for the affine equalities and the objective. Hence $\alpha(u, v, t) + (1-\alpha)(u', v', t') \in \mathcal{G}$.

Step 2: The point $(0, 0, p^*)$ is on the boundary of $\mathcal{G}$. By definition of p^*, the point $(0, 0, p^*)$ lies in $\operatorname{cl}(\mathcal{G})$ (it is achieved in the limit by a minimizing sequence). However, $(0, 0, p^* - \epsilon) \notin \mathcal{G}$ for any $\epsilon > 0$, since that would require a feasible x with $f_0(x) \leq p^* - \epsilon < p^*$, contradicting the definition of p^*.

Step 3: Supporting hyperplane. By the supporting hyperplane theorem for convex sets, there exists $(\tilde{\lambda}, \tilde{\nu}, \mu) \neq (0, 0, 0)$ with $\tilde{\lambda} \in \mathbb{R}^m$, $\tilde{\nu} \in \mathbb{R}^p$, $\mu \in \mathbb{R}$ such that

$$\tilde{\lambda}^\top u + \tilde{\nu}^\top v + \mu\, t \geq \mu\, p^* \tag{12}$$

for all $(u, v, t) \in \mathcal{G}$. Since $(u, v, t) \in \mathcal{G}$ allows $u_i \to +\infty$, we must have $\tilde{\lambda}_i \geq 0$ for each i; and since $t \to +\infty$ is allowed, $\mu \geq 0$.

Step 4: Show $\mu > 0$ using Slater's condition. Suppose for contradiction that $\mu = 0$. Then (12) becomes $\tilde{\lambda}^\top u + \tilde{\nu}^\top v \geq 0$ for all $(u, v, t) \in \mathcal{G}$. Evaluate at the Slater point $\hat{x}$: take $u_i = f_i(\hat{x}) < 0$, $v_j = 0$, $t = f_0(\hat{x})$. This gives

$$\sum_{i=1}^{m} \tilde{\lambda}_i f_i(\hat{x}) \geq 0.$$

Since $\tilde{\lambda}_i \geq 0$ and $f_i(\hat{x}) < 0$, each term is non-positive. The sum being non-negative forces every term to be zero. Since $f_i(\hat{x}) < 0$, we get $\tilde{\lambda}_i = 0$ for all i.

Now the inequality reduces to $\tilde{\nu}^\top v \geq 0$ for all achievable v. Since the equality constraints are affine, $v = Ax - b$ for a matrix A with rows $a_j^\top$ and vector b with entries b_j. As x ranges over $\mathbb{R}^n$, v ranges over $\text{range}(A) - b \supseteq \{0\}$. For any direction $d \in \text{range}(A)$, both d and $-d$ are achievable perturbations, so $\tilde{\nu}^\top d \geq 0$ and $-\tilde{\nu}^\top d \geq 0$, giving $\tilde{\nu}^\top d = 0$ for all d in the range. This requires $\tilde{\nu} = 0$ (assuming the equality constraints are non-redundant; otherwise $\tilde{\nu}$ lies in the null space of $A^\top$, but the supporting hyperplane must be nonzero). This contradicts $(\tilde{\lambda}, \tilde{\nu}, \mu) \neq (0, 0, 0)$.

Therefore $\mu > 0$.

Step 5: Conclude strong duality. Divide (12) by $\mu > 0$ and define $\lambda^* = \tilde{\lambda}/\mu$, $\nu^* = \tilde{\nu}/\mu$. We have $\lambda^* \geq 0$. For any x that achieves (u, v, t) with $u_i = f_i(x)$, $v_j = a_j^\top x - b_j$, $t = f_0(x)$:

$$\sum_{i=1}^{m} \lambda_i^* f_i(x) + \sum_{j=1}^{p} \nu_j^* (a_j^\top x - b_j) + f_0(x) \geq p^*.$$

Hence $L(x, \lambda^*, \nu^*) \geq p^*$ for all x. Taking the infimum over x:

$$g(\lambda^*, \nu^*) = \inf_x L(x, \lambda^*, \nu^*) \geq p^*.$$

By weak duality (Theorem 3.2), $g(\lambda^*, \nu^*) \leq p^*$. Therefore $g(\lambda^*, \nu^*) = p^*$, so $d^* \geq g(\lambda^*, \nu^*) = p^*$. Combined with $d^* \leq p^*$ from weak duality, we conclude $d^* = p^*$, and the dual optimum is attained at (λ^*, ν^*). □

Strong Duality via Slater's Condition

For a convex program, if a strictly feasible point exists (Slater's condition), then $d^* = p^*$. The dual optimum is attained. This converts the constrained primal problem into an equivalent (unconstrained concave) dual problem.

5 Karush–Kuhn–Tucker Conditions

Definition 5.1 (KKT conditions)**.** For the convex program (1) with differentiable $f_0, f_1, \ldots, f_m$, the *Karush–Kuhn–Tucker (KKT) conditions* for a primal-dual triple (x^*, λ^*, ν^*) are:

(KKT1) Primal feasibility: $f_i(x^*) \leq 0$ for $i = 1, \ldots, m$; $\quad a_j^\top x^* = b_j$ for $j = 1, \ldots, p$.

(KKT2) Dual feasibility: $\lambda_i^* \geq 0$ for $i = 1, \ldots, m$.

(KKT3) Complementary slackness: $\lambda_i^* f_i(x^*) = 0$ for $i = 1, \ldots, m$.

(KKT4) Stationarity: $\nabla f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^{p} \nu_j^* a_j = 0.$

Remark 5.2 (Interpretation of complementary slackness). The condition $\lambda_i^* f_i(x^*) = 0$ says: for each inequality constraint, either the constraint is *active* ($f_i(x^*) = 0$) or its multiplier is zero ($\lambda_i^* = 0$), or both. A multiplier $\lambda_i^* > 0$ means the constraint is "binding"—removing it would improve the objective.

Theorem 5.3 (KKT necessary conditions)**.** *Suppose strong duality holds for* (1)*, the functions* $f_0, f_1, \ldots, f_m$ *are differentiable,* x^* *is primal optimal, and* (λ^*, ν^*) *is dual optimal. Then the KKT conditions hold at* (x^*, λ^*, ν^*)*.*

Proof. Since strong duality holds:

$$f_0(x^*) = p^* = d^* = g(\lambda^*, \nu^*) = \inf_x L(x, \lambda^*, \nu^*) \leq L(x^*, \lambda^*, \nu^*). \tag{13}$$

Expanding the Lagrangian at (x^*, λ^*, ν^*):

$$L(x^*, \lambda^*, \nu^*) = f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* f_i(x^*) + \sum_{j=1}^{p} \nu_j^* \underbrace{(a_j^\top x^* - b_j)}_{=0}. \tag{14}$$

The equality constraint terms vanish by primal feasibility. The inequality constraint terms satisfy $\lambda_i^* f_i(x^*) \leq 0$ (since $\lambda_i^* \geq 0$ and $f_i(x^*) \leq 0$). Therefore:

$$L(x^*, \lambda^*, \nu^*) = f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* f_i(x^*) \leq f_0(x^*). \tag{15}$$

Combining (13) and (15):

$$f_0(x^*) \leq L(x^*, \lambda^*, \nu^*) \leq f_0(x^*),$$

so $L(x^*, \lambda^*, \nu^*) = f_0(x^*)$. This means $\sum_{i=1}^{m} \lambda_i^* f_i(x^*) = 0$. Since each term $\lambda_i^* f_i(x^*)$ is non-positive, each must be zero:

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \ldots, m. \tag{16}$$

This establishes **complementary slackness**.

From (13), $g(\lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*)$, which means x^* attains the infimum in the definition of $g(\lambda^*, \nu^*)$. Since $x \mapsto L(x, \lambda^*, \nu^*)$ is a differentiable convex function (it is a non-negatively weighted sum of convex functions) and x^* is its global minimizer over $\mathbb{R}^n$, the gradient must vanish:

$$\nabla_x L(x^*, \lambda^*, \nu^*) = \nabla f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^{p} \nu_j^* a_j = 0. \tag{17}$$

This establishes **stationarity**. Primal and dual feasibility hold by assumption. □

Theorem 5.4 (KKT sufficient conditions for convex problems)**.** *For the convex program* (1) *with differentiable* $f_0, f_1, \ldots, f_m$*, if a triple* (x^*, λ^*, ν^*) *satisfies the KKT conditions, then* x^* *is primal optimal and* (λ^*, ν^*) *is dual optimal, with zero duality gap.*

Proof. **Step 1:** By stationarity (KKT4), $\nabla_x L(x^*, \lambda^*, \nu^*) = 0$. The function $x \mapsto L(x, \lambda^*, \nu^*)$ is convex in x (since $f_0, f_1, \ldots, f_m$ are convex, $\lambda^* \geq 0$, and the remaining terms are affine). A differentiable convex function with vanishing gradient at x^* attains its global minimum there. Hence:

$$g(\lambda^*, \nu^*) = \inf_x L(x, \lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*). \tag{18}$$

Step 2: By complementary slackness (KKT3) and primal feasibility (KKT1):

$$L(x^*, \lambda^*, \nu^*) = f_0(x^*) + \underbrace{\sum_{i=1}^{m} \lambda_i^* f_i(x^*)}_{=0} + \underbrace{\sum_{j=1}^{p} \nu_j^* (a_j^\top x^* - b_j)}_{=0} = f_0(x^*). \tag{19}$$

Step 3: Combining: $g(\lambda^*, \nu^*) = f_0(x^*)$. By weak duality (Theorem 3.2):

$$d^* \geq g(\lambda^*, \nu^*) = f_0(x^*) \geq p^* \geq d^*.$$

All inequalities are equalities. Hence $f_0(x^*) = p^*$ (x^* is primal optimal), $g(\lambda^*, \nu^*) = d^*$ ((λ^*, ν^*) is dual optimal), and $p^* = d^*$ (zero duality gap). □

KKT Conditions for Convex Problems

For differentiable convex programs, the KKT conditions are:

- **Necessary** for optimality, provided a constraint qualification (such as Slater's condition) holds.
- **Sufficient** for optimality, without any constraint qualification.

Together, the KKT conditions provide a complete characterization of primal-dual optimal pairs.

RL/ML Connection

Constrained MDPs. In safe reinforcement learning, the agent maximizes expected return subject to constraints on expected cost: $\max_\pi J(\pi)$ s.t. $C_i(\pi) \le d_i$. The Lagrangian relaxation is $\max_\pi \min_{\lambda \ge 0}\{J(\pi) - \sum_i \lambda_i(C_i(\pi) - d_i)\}$. Slater's condition guarantees strong duality when a strictly feasible policy exists, justifying primal-dual algorithms such as Lagrangian PPO.
Trust-region methods. In TRPO, the policy update solves $\max_\theta L(\theta)$ s.t. $\mathrm{KL}(\pi_{\text{old}} \| \pi_\theta) \le \delta$. The KKT multiplier λ^* on the KL constraint determines the effective step size. Complementary slackness says: either the KL constraint is active (the update reaches the trust-region boundary) or $\lambda^* = 0$ (the unconstrained maximum lies inside the trust region).

6 Detailed Examples

Example 6.1 (Water-filling)**.** Consider the problem of allocating power across n channels to maximize total capacity:

$$\max_{x \ge 0} \sum_{i=1}^{n} \log(\alpha_i + x_i) \quad \text{s.t.} \quad \sum_{i=1}^{n} x_i = P,$$

where $\alpha_i > 0$ are channel gains and $P > 0$ is total power. Equivalently, minimize $-\sum_i \log(\alpha_i + x_i)$. The Lagrangian is

$$L(x, \lambda, \nu) = -\sum_i \log(\alpha_i + x_i) + \sum_i \lambda_i(-x_i) + \nu\Big(\sum_i x_i - P\Big).$$

The KKT stationarity condition gives $\frac{-1}{\alpha_i + x_i^*} - \lambda_i^* + \nu^* = 0$, hence $x_i^* + \alpha_i = \frac{1}{\nu^* - \lambda_i^*}$. Complementary slackness ($\lambda_i^* x_i^* = 0$) yields the "water-filling" solution:

$$x_i^* = \max\Big(\frac{1}{\nu^*} - \alpha_i,\ 0\Big),$$

where ν^* (the "water level") is chosen so that $\sum_i x_i^* = P$.

Example 6.2 (Equality-constrained QP via KKT)**.** Consider $\min \frac{1}{2}x^\top P x + q^\top x$ s.t. $Ax = b$ with $P \succ 0$. There are no inequality constraints, so the KKT conditions reduce to stationarity and primal feasibility:

$$Px^* + q + A^\top \nu^* = 0, \qquad Ax^* = b.$$

In matrix form: $\begin{pmatrix} P & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} x^* \\ \nu^* \end{pmatrix} = \begin{pmatrix} -q \\ b \end{pmatrix}$. This is the *KKT system*, a linear system of equations.

Example 6.3 (Sensitivity interpretation)**.** Consider $p^*(\epsilon) = \min\{f_0(x) : f_1(x) \le \epsilon\}$. If strong duality holds and λ^* is the optimal dual variable, then $\frac{dp^*}{d\epsilon}\big|_{\epsilon=0} = -\lambda^*$. A large multiplier means the constraint is "expensive": relaxing it slightly yields a large improvement in the objective.

7 Exercises

Exercise 7.1 (Weak duality bound)**.** Consider $\min x_1 + x_2$ s.t. $x_1^2 + x_2^2 \le 1$. Find the dual function $g(\lambda)$, evaluate it at $\lambda = 1$, and verify that $g(1) \le p^*$. Then find d^* and verify strong duality.

Exercise 7.2 (Slater verification)**.** For the problem $\min e^{-x}$ s.t. $x^2/y \le 0$, $y > 0$, determine whether Slater's condition holds. If not, find the duality gap.

Exercise 7.3 (KKT for box constraints)**.** Solve $\min (x_1 - 3)^2 + (x_2 - 5)^2$ s.t. $0 \le x_1 \le 4$, $0 \le x_2 \le 3$ by writing and solving the KKT conditions. Identify which constraints are active and which multipliers are nonzero.

Exercise 7.4 (Dual of norm minimization)**.** Derive the dual of the problem $\min \|x\|_2$ s.t. $Ax = b$. Show that the dual is $\max b^\top \nu$ s.t. $\|A^\top \nu\|_2 \le 1$. Verify that strong duality holds (under what condition?).

Exercise 7.5 (Complementary slackness in LP)**.** Given the LP $\min c^\top x$ s.t. $Ax \le b$, $x \ge 0$ and its dual $\max b^\top y$ s.t. $A^\top y + s = c$, $y, s \ge 0$, show that at optimality $x_i^* s_i^* = 0$ and $y_j^*(b_j - A_j x^*) = 0$. Explain the economic interpretation: each variable is either zero or its associated constraint is binding.

Exercise 7.6 (CMDP Lagrangian relaxation)**.** Consider a constrained MDP: $\max_\pi \mathbb{E}[\sum_t \gamma^t r_t]$ s.t. $\mathbb{E}[\sum_t \gamma^t c_t] \le d$. Write the Lagrangian. Show that for fixed $\lambda \ge 0$, the inner maximization over π reduces to an unconstrained MDP with modified reward $r_t + \lambda(d - c_t)$. Explain how to find the optimal λ^* via a one-dimensional search.

Exercise 7.7 (Non-convex duality gap)**.** Consider $\min x^2$ s.t. $(x-1)(x-2) \le 0$ (feasible set $[1, 2]$). Compute p^*, derive the dual function, compute d^*, and show that there is a nonzero duality gap. Explain why Slater's condition is not applicable.

References